Setting up Integrated Ciruit Layout Design Software

IIC_OSIC_TOOLS first setup and launch

IIC tools git is available under this link. The whole setup process comes down to one-line tool setup and installing Docker.

1. If you dont have curl, install it.

```
sudo apt install curl
```

2. Then install IIC tools, the one-line setup is:

```
curl -fsSL https://osic.tools/install.sh | bash
```

3. Add new user to docker.

```
sudo usermod -aG docker $USER
```

4. Apply changes to your current terminal session.

```
newgrp docker
```

5. If you dont want to repeat this step each time you login, you should reboot the system.

```
sudo reboot
```

6. Then follow the Docker installation process described on their site
7. There are few ways to launch the tool. firstly go to the iic-osic-tools folder in your home directory. then you can launch the enviroment by using commands:
   `./start_x.sh` - launches the x
   `./start_vnc.sh` - x
   `./start_shell.sh` - x
   Just keep in mind that first launch might take a few minutes before the program launches.

Getting familiar with IIC-OSIC-TOOLS and sky130A library

This guide will follow the local enviroment started up with `./start_x.sh`. It should launch a terminal.

1. With command `sak-pdk` you can switch between different PDKs, type this command to see the usage. In this tool, there are 4 available by default PDKs with them being:
    - `sky130A` - SkyWater Technologies, 180-130nm hybrid 1.8-3.3V
    - `gf180mcuD` - Global Foundries, 180nm 3.3-6V
    - `ihp-sg13g2` - IHP Microelectronics
    - `ihp-sg13cmos5l` - IHP Microelectronics
2. This instruction will get you familiar with sky130A PDK so use `sak-pdk sky130A`. If this command is not input before starting your work, you will be using default PDK, which is `ihp-sg13g2`. You can check current PDK with `echo $PDK` after launch.
3. Then you can launch schematic capture and netlisting EDA tool - Xschem with command:

```
xschem [filename].sch
```

4. Feel free to explore Menu Bar, especially the `Help -> Keys/Keybindings` that has many useful shortcuts and `Help -> About XSCHEM` from which can check program's git or [Xschem Manual].
5. x